



LINAC ACIS Upgrade Project Plan

1.12.52.1- Rev 0

Date: 2008-09-25

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Revision History

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1.0 OVERVIEW

1.1 PROJECT GOALS

The existing LINAC system makes use of a Micro84 based ACIS system. Given the age of the system, difficulty in procurement of spares (since the vendor has discontinued support for the platform) it now needs to be replaced. This will ensure the LINAC ACIS to be consistent with the current ACIS present used by beamlines, booster ring, and storage ring.

The system shall be replaced with a form-fit-function-replacement with the following exceptions:

- The LINAC power supply room (Room 0102) will be removed from the LINAC ACIS system (with adequate adjustments to zone entrances)
- The elevator stairwell will be removed from the LINAC ACIS system.
- Additional emergency off buttons will be added if required

1.2 ACRONYMS

ACIS	Access Control Interlock System
CID	Control & Instrumentation Development
ETS	Engineering & Technical Service
HSE	Health, Safety & Environment
LINAC	Linear Accelerator

1.3 PROJECT GOVERNANCE

Canadian Light Source Inc. (CLSI) is a legally incorporated entity. CLSI and the CLS facility are owned by the University of Saskatchewan. The relationship between CLSI and the University of Saskatchewan is defined in a license agreement between the two organizations. This license agreement covers the activities associated with operating the CLSI facility including its beamlines. CLSI retains all responsibility for health, safety, environment and design authority over the facility. CLSI holds a Particle Accelerator Operating Licence (PA1OL-02.00/2012) issued by the Canadian Nuclear Safety Commission to operate the CLS facility. The license agreement between CLSI and the University of Saskatchewan is included in Appendix A of the CLSI operating license t. Detailed line management responsibility and authority is defined in [1].

Key Roles:

HSE Manager	- Management of verification & acceptance of system
HSE RPC Coordinator/HSE Technician	- Verification & Validation of the System
CID Manager	- Management oversight of design/installation activates
CID Project Lead	- Design Development and Direct Installation

2.0 STANDARDS AND TECHNOLOGY

The LINAC ACIS system will use the same architecture and technology as the one currently used on other BR1 and SR1 tunnels [2]. The same hardware and software development techniques already used for safety critical system and software design at the CLS will be applied to the LINAC ACIS system. The design is based on IEC 61508 [3] certified control equipment following TÜV application guidelines [4] and the PLC vendor guidelines [5].

2.1 STANDARDS

- System Engineering Guide [7]
- CLS Safety System Development Strategy [8]
- Quality Assurance Manual [9]
- Canadian Electrical Code¹.

2.2 VENDOR STANDARDS

- Refer to Siemens SIMATIC S7-400F and S7-400FH Fail-Safe Systems Manual [5], and SIMATIC Automation System S7-300 Fail-Safe Signal Module [6].

3.0 WORK BREAKDOWN STRUCTURE

		Department	Timeline
1.0	Inception		
1.1	Radiation Level Measurement	HSE	Spring 08
1.2	Hazard Analysis	HSE	June 08
1.3	Radiation Analysis	HSE	September 08
2.0			
2.1	Requirement Document	CID	June 08
2.2	Lockup Zone Plan	HSE/CAD	July 08
2.3	ACIS Zone Layout Detail	HSE/CAD	July 08
3.0	Design (Elaboration)		
3.1	Control	CID	September 08
3.2	Civil (Door etc.)	ETS	October 08
3.3	V&V Procedure	CID/HSE	October 08
3.4	Lockup Procedure	HSE	September 08
4.0	CNSC Approval		
4.1	Safety Report	HSE	October 08
4.2	Application Letter and associated documents		October 18, 2008
4.3	CLSI CMD		December 18, 2008

¹ As per the Electrical Inspections Act of Saskatchewan E-6.3

4.4	Public Hearing		February 18, 2009
4.5	CNSC Decision		April 06, 2009
5.0	Installation	CID	April, 2009
6.0	Verification and Validation	HSE	April, 2009
7.0	Close-Out		

Appendix A is a proposed timeline for activities from Hazard Analysis to CNSC Review, which covers a timeframe from middle June to October, 2008. Note that the milestone dates on the timeline are based on best estimate, since there is no historical data available to make a more accurate estimate. The value of this diagram is that it lays out the things need to be done before Oct 1st and the right sequence of doing them.

4.0 BUDGET

4.1 MATERIAL AND SUPPLIES

PLC Equipment	\$100,000
Conduit and Cabling	Budget 50% of PLC Equipment
Doors and Structural Modifications	\$100,000
Hardwire Chain	\$50,000

4.2 LABOUR

TBD

5.0 DOCUMENTATION PLAN

Based on the CNSC requirements [10], the following documents should be submitted.

Document	Prepared By	Reviewed By	Approved By
Project Planning			
LINAC ACIS Upgrade Project Plan (1.12.52.1) – New	H. Zhang G. Cubbon	R. Tanner	E. Matias M Benmerrouche
Design Input and Safety Analysis Documentation			
Safety System Development Strategy (0.2.37.2) – Minor Modification	E. Matias	H. Zhang M. Benmerrouche M. de Jong	J. Hormes
CLS Safety Report (11.18.40.2)	E. Matias M. Benmerrouche	H. Zhang M. Benmerrouche M. de Jong	J. Hormes
LINAC ACIS Upgrade Hazard	G. Cubbon	A. Hodges	M. Benmerrouche

and Risk Analysis (11.18.52.1) - New		R. Tanner H. Zhang	
Routine Operation Limits and Conditions (8.1.1.6)	No change currently expected.		
Human Factors Workscope (0.1.1.1)	No change currently expected.		
LINAC ACIS Upgrade Radiation Shielding Analysis (11.18.53.19) – New	G. Cubbon.	L. Cowles, S. Carrere	M. Benmerrouche
Design Documents			
LINAC ACIS Design Manual (Note: this manual includes the requirement) (1.9.52.1) – New	H. Zhang	G. Cubbon. R. Tanner	E. Matias
Booster/Storage Ring/Beamlines ACISPLC Component Manual (7.9.39.4)	No change currently expected.		
Personnel Safety Lockup System (1.2.37.2)	(Canceled Replaced by 1.9.52.1)		
Personal Safety Lockup System Micro84 Program	(Canceled Replaced by 1.9.52.1)		
CLS Lockup PLC Technical Specification (7.4.37.1)	No change currently expected.		
LINAC Lockup Zone Plan Master Drawing	G. Judkins	G. Cubbon	M. Benmerrouche
LINAC ACIS Layout & Component Drawings	D. Bergstrom	G. Cubbon	E. Matias
LINAC ACIS Wiring & Panel/Rack Layout Drawings	D. Bergstrom	H. Zhang	E. Matias
User Procedures and Training			
LINAC ACIS Lockup Procedure (2.7.37.2) – Revise	G. Cubbon	L. Cowles, A. Hodges	M. Benmerrouche
Accelerator ACIS Lockup Training (11.14.52.5) – Minor Change	G. Cubbon	M. Silzer A. Hodges	M. Benmerrouche
ACIS Accelerator Lockup Training Practical Session Instructor Checklist (11.11.52.10)	No change currently expected		
Testing, Verification and Validation Procedures			
LINAC ACIS V&V Procedure -	C. Finlay	G. Cubbon H. Zhang	M. Benmerrouche

LINAC ACIS Human Factors Validation Procedure	C. Finlay	G. Cubbon M. McKibben	E. Matias
Safety System Ringout Procedure (7.7.52.1)	No change currently expected.		

6.0 HUMAN FACTORS

It is expected the operator interface for the new ACIS system will be very similar to the existing system. The system will be designed in accordance with the CLS Human Factors Workscope [11] and a LINAC ACIS specific Human Factors Validation procedure will be developed.

7.0 REFERENCES

- [1] Fetch, N. 2004. Organization Management. CLS Doc. No. 0.1.1.15 Rev. 0.
- [2] Tanner, R. 2005. Booster Storage Ring Beamline Access Control and Interlock System. CLS Doc No. 7.9.39.4 Rev. 3.
- [3] IEC 61508-0 2005. Functional Safety of electrical/electronic/programmable electronic safety-related systems Part 0 – Functional safety and IEC 61508. International Electrotechnical Commission. 2005.
- [4] TÜV 2000. Report on Certificate Z2 99 12 38282 001 Safety Related Programmable Systems SIMATIC S7-400F and S7-400FH Report No. 10042360. TÜV Product Services GmbH, München Germany
- [5] Siemens. 2000. SIMATIC S7-400F and S7-400FH Fail-Safe Systems Manual. Document A5E00085588-02 Part No. 6ES7988-8FA10-8BA0. Siemens AG. Nuernberg, Germany.
- [6] Siemens. 2002. SIMATIC Automation System S7-300 Fail-Safe Signal Module. Document A5E00137903-03. Part No. 6ES7988-8FA10-8BA0. Siemens AG. Nuernberg, Germany.
- [7] Matias, E. 2007a. System Engineering Guide. CLS Doc. No. 0.1.69.1 Rev. 1.
- [8] Matias, E. 2007b. CLS Safety System Development Strategy. CLS Doc. No. 0.2.37.2. Rev. 2.
- [9] Matias, E. 2007. Quality Assurance Manual. CLS Doc. No. 10.12.1.1 Rev. 6
- [10] Letter from Jonathan Schmidt (CNSC) to M. Benmerrouche (CLSI), “Upgrade to LINAC Access Control Interlock System (ACIS)”, January 12, 2007 (CNSC file Notre reference 29-1-2-4-0)
- [11] McKibben, M. 2008. “CLS Human Factors Workscope”, 0.1.1.1, Rev. 1

Appendix A: Project Timeline (June 2008 – October 2008))

